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Integrated ESP Wells Platform for Optimization and Management Based on Mechanism Analysis and Data-Driven Model

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Abstract

The optimization and management for ESP wells is an integrated engineering technology. Only if the oil field have complete data of Geological, Reservoir, Wellbore, ESP, and even electrical, the ESP's production status can be accurate judgment. So, there is a great demand of an integrated method to diagnose faults, analyze fault causes, and propose optimization measures. Then, using an integrated platform to visualize production data and analysis results. In this study, a mechanism analysis logic has been built first based on the onsite experience and integrated database including Geological, Reservoir, Wellbore, and ESP data. Then, different data type uses different machine learning method. For example, the vibration analysis using frequency/time domain, the Current card using image recognition (CNN), the pressure and production data using Random Forest. After that, using a neural network to combine the mechanism analysis logic and all the machine learning models to improve the accuracy. The platform is built to data and results visualization including three modules: Data monitor, Process automation, Rapid abnormal response. The model in this platform includes both experience, mechanism, and data-driven, with higher reliability, controllability, and interpretability. Moreover, the model has a full process function. By inputting the initial dataset, it can quickly respond to the diagnosis results of ESP faults and then corresponding process optimization measures. By using such a platform, on-site work can be greatly simplified, and the management of ESP wells can be more standardized and modular. With the automatic data collection and real-time transmission, the platform can diagnose the current production status of ESP wells in real time, warn of possible future

failures, and propose trouble shooting measures. The visualized ESP management platform integrates data-driven methods, with built-in algorithms for automatic updating, achieving self-iterative updates and upgrades, making the management of ESP wells more efficient. After the deployment of the oil field, the model will continue to optimize and improve based on real-time data, which can effectively increase the MTBF (Mean time before faults) of the ESP.

Keywords: Intelligent management platform, Fault diagnose, Early warning, Production optimization

Introduction

Electric Submersible Pumps (ESPs) are critical artificial lift systems in oilfield production, enabling efficient extraction of oil from low pressure reservoirs^{[1][2]}. However, optimizing and managing ESP operations remains a complex engineering challenge due to the interdependence of geological, reservoir, wellbore, and equipment parameters^{[3][4][5]}. Traditional approaches often rely on fragmented data analysis, manual diagnostics, and heuristic experience, leading to delayed fault detection, suboptimal performance, and costly downtime. With the increasing complexity of mature oilfields and the demand for cost-effective production, there is an urgent need for integrated solutions that combine mechanistic understanding with advanced data-driven methodologies to achieve real-time monitoring, predictive maintenance, and intelligent decision-making.

Current ESP management systems face three primary limitations^{[6][7]}: (1) Disconnected data ecosystems, where geological, operational, and equipment parameters are stored in siloed databases, hindering holistic analysis; (2) Overreliance on threshold-based alarms and manual diagnostics, which lack adaptability to dynamic well conditions and fail to uncover root causes; (3) Inadequate integration of physical models with machine learning techniques, resulting in poor interpretability and delayed response to emerging failures. While data-driven models such as convolutional neural networks (CNNs) and random forests have shown promise in fault detection, their standalone application often overlooks critical domain knowledge, limiting accuracy and practicality in field deployments^{[8][9][10]}.

This study addresses these gaps by introducing an integrated ESP wells' platform that synergies mechanistic analysis with hybrid machine learning architectures. The platform establishes a unified data framework spanning geological, reservoir, wellbore, and ESP systems, enabling multidimensional correlation of static and dynamic parameters. A novel diagnostic framework is developed, combining physics-based logic tables for root cause analysis with specialized machine learning models—CNN for current card pattern recognition, isolation forests for vibration anomaly detection. By embedding real-time predictive capabilities and 3D visualization tools, the system not only identifies existing faults but also forecasts equipment degradation trends and optimizes production parameters.

Data automatic collection method

By establishing a multidimensional monitoring system through sensor networks to collect real-time parameters such as wellhead pressure, liquid level, temperature, motor current, motor voltage, and frequency, combined with dynamic data like inlet/outlet pressure differentials and liquid production rates^{[11][12]}. Current ESPs are equipped with real-time parameter acquisition modules that continuously monitor essential parameters including pump current, voltage, discharge pressure, intake pressure, pump vibration, and motor temperature. For individual-well production dynamics, real-time metering of multiple phases (oil, gas, and water) at the wellhead remains a persistent challenge in oilfield production management^{[13][14][15]}.

The flow patterns of three-phase (gas-liquid-solid) systems are influenced by factors such as flow velocity, gas-liquid ratio, and phase-specific properties (e.g., density, viscosity, surface tension), potentially forming diverse regimes including laminar flow, bubble flow, slug flow, and mist flow. In contrast, gas-

liquid two-phase flows—though also exhibiting regime transitions (e.g., bubble-to-annular flow)—involve fewer parametric dimensions due to the simplified interaction between gas and liquid phases. The mixing behavior of oil, gas, and water in three-phase flow systems demonstrates significantly greater complexity, with substantial velocity differentials between phases that lead to increased susceptibility to sensor signal interference, making it particularly challenging to establish a test model.

Oil-water-gas three-phase metering devices can be categorized into two types based on the metering method: separation metering and non-separation metering (Fig. 1). Separation metering involves physically separating the phases (e.g., gas-liquid separation, oil-water settling) before measuring the flow rate of each phase individually. This avoids interference from the complex flow patterns of unseparated mixed fluids, resulting in relatively higher accuracy. Although skid-mounted designs are possible, the overall system remains bulky and costly. Additionally, it requires waiting for physical separation, making real-time continuous measurement impossible, and is less adaptable for single-well metering. Non-separation metering employs methods such as microwaves or radiation for measurement. It is more compact and can operate without intruding into the flow path, making it suitable for real-time continuous single-well metering. However, its accuracy varies significantly and is easily affected by fluid properties and testing conditions. To ensure measurement accuracy, the most common approach currently is skid-mounted separation metering, where one metering skid serves 3-5 wells, switching between wells at different time intervals for separated metering. Alternatively, to balance real-time measurement with accuracy, another method involves using real-time two-phase online flow measurement to determine gas and liquid production rates, followed by periodic sampling of liquid volumes to measure water cut via separation. In this approach, gas and total liquid production data are real-time, while water and oil production rates are averaged over specific time intervals.

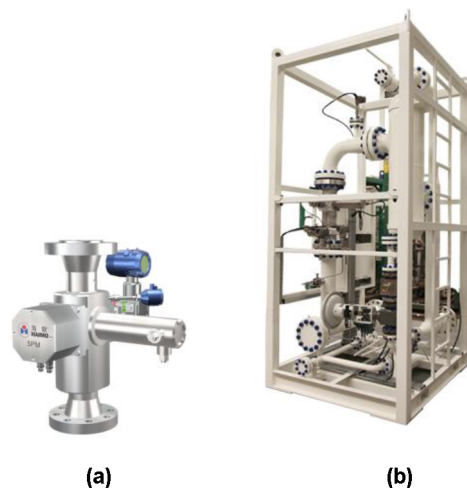


Figure 1—HAIMO flow meter device: (a) Separation metering device; (b) Non-separation metering device.

Management of integrated data

The platform's data management system is built on a professional database, enabling standardized storage and integrated management of single-well data across the entire oilfield. To ensure the accuracy of production status analysis for electric submersible pump (ESP) wells, a multi-dimensional data framework covering geology-reservoir-wellbore-ESP systems must be established, as shown in Fig. 2. Specifically^[16]:

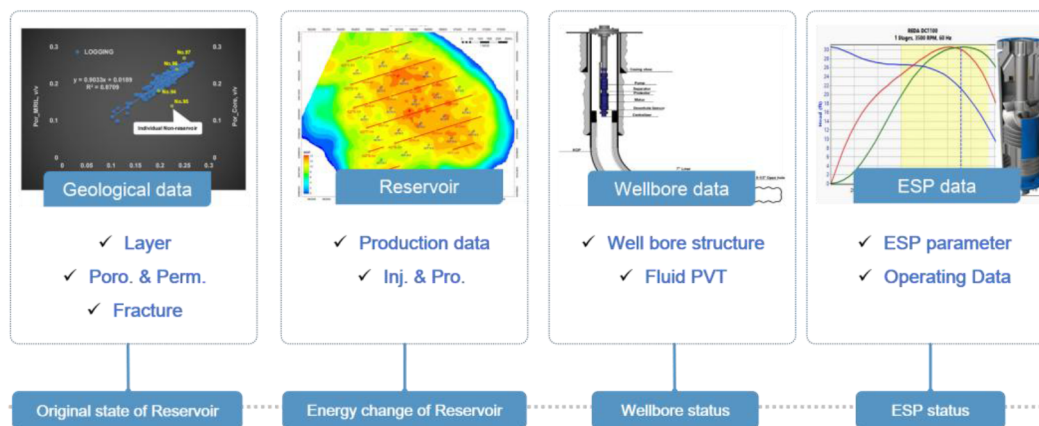


Figure 2—Integrated database

Reservoir data: Includes reservoir parameters such as producing zones, average permeability, porosity, as well as production performance data like historical production rates and pressure trends.

Wellbore data: Covers key parameters such as well structure, completion methods, and fluid properties.

ESP data: Involves operational information like pump setting depth, workover records, and failure characteristics.

Based on data update characteristics, the data can be categorized into:

Static data: Relatively stable foundational data, such as porosity-permeability parameters, producing zones, and well structure.

Dynamic data: Time-varying monitoring data, including real-time production rates, formation pressure, and ESP operating parameters (current, intake/discharge pressure, vibration, etc.).

Data analysis for diagnose and warning

Diagnose method

Fault diagnosis in ESP wells is a critical technology for ensuring efficient oil and gas production. Its core lies in integrating static well data with real-time monitoring data, employing mechanistic analysis combined with intelligent data analytic to swiftly identify fault types and uncover their root causes.

Based on the influence patterns of ESP parameters and well production parameters under different operating conditions, combined with field production experience, the integrated mechanistic analysis logic table has been built. This table including fault diagnosis, root cause analysis, and production optimization recommendations. By utilizing this comprehensive logic analysis table in conjunction with both dynamic and static well parameters, the system achieves rapid fault diagnosis, efficient cause identification, and timely production optimization([Table 1](#)).

Table 1—Integrated mechanistic analysis logic table

Optimization method	Cause	Faults	ESP operating data							Production data					ESP parameter			Well parameter		
			motor current	motor Tem	voltage	Intake pressure	Discharge pressure	vibration	Running time	Tubing pressure	Casing pressure	Liquid rate	GFR	WC	Frequency	Max rate	Min rate	Density	Viscosity	Asphaltene content
Demulsification	Block by emulsification	Block	Increase	Increase	Stable	Increase	Decrease	Stable	/	Decrease	Stable	Decrease	Stable	Increase	Stable	/	/	Increase	Increase	Stable
Well washing	Block by impurity	blockage	Increase	Increase	Stable	Increase	Decrease	Stable	/	Decrease	Stable	Decrease	Stable	Stable	Stable	/	/	Stable	Stable	Stable
Add dissolving agent	Blockage by asphaltene/scale	blockage	Increase	Increase	Stable	Increase	Decrease	Stable	/	Decrease	Stable	Decrease	Stable	Stable	Stable	/	/	Increase	Increase	Increase
Optimize injection	lack injection	Underload	Decrease	Increase	Stable	Decrease	Stable	Stable	/	Decrease	Stable	Decrease	Increase	Decrease	Stable	/	/	Decrease	Stable	Stable
Acid/Optimize injection	Reservoir capacity decrease	Underload	Decrease	Increase	Stable	Decrease	Stable	Stable	/	Decrease	Stable	Decrease than design	Stable	Stable	Stable	/	/	Stable	Stable	Stable
Change pump p/increase choke size	Unsuitable pump design	Underload	Decrease	Increase	Stable	Decrease	Stable	Stable	/	Decrease	Stable	Decrease than design	Stable	Stable	Stable	/	/	Stable	Stable	Stable
Optimize frequency	Pump frequency is too high	Underload	Decrease	Increase	Stable	Decrease	Increase	Stable	/	Increase	Stable	Increase	Stable	Stable	Higher than design	/	/	Stable	Stable	Stable
Stop pump		Load fluctuation	Continued volatility	Stable	Wave	Wave	Wave	Stable	/	Wave	Stable	Wave	Stable	Stable	Stable	/	Bigger than Pro	Stable	Stable	Stable
Optimize injection	High injection	Overload	Increase	Stable	Stable	Increase	Increase	Stable	/	Increase	Stable	Increase	/	/	Stable	/	/	Stable	Stable	Stable
Change pump p/adjust choke size	Unsuitable pump style	Overload	Increase	Stable	Stable	Increase	Increase	Stable	/	Increase	Stable	Increase than design	Stable	Stable	Stable	Smaller than Pro	/	Stable	Stable	Stable
Optimize frequency	lower frequency	Overload	Increase	Stable	Stable	Increase	Increase	Stable	/	Increase	Stable	Increase than design	Stable	Stable	Smaller than design	/	/	Stable	Stable	Stable
Demulsifier	Emulsification	Overload	Increase	Stable	Stable	Increase	Increase	Stable	/	Increase	Stable	Decrease	/	Increase	Stable	/	/	Increase	Increase	Stable
Change pump p	Heavy oil Increase	Overload	Increase	Stable	Stable	Increase	Increase	Stable	/	Increase	Stable	Decrease	Stable	Stable	Stable	/	/	Increase	Increase	Stable/Increase
	Mechanical failure (insulation, protective cover)	Overcurrent	Sudden increase trigger protection	Stable	Decrease	Decrease	Decrease	Stable	/	Decrease	Stable	Decrease to 0	Stable	Stable	Stable	/	/	Stable	Stable	Stable
	lack voltage	lake pressure	Decrease	Stable	Decrease	Decrease	Decrease	Stable	/	Decrease	Stable	Decrease	Stable	Stable	Stable	/	/	Stable	Stable	Stable
Well washing	Fluid impurities	Wave-normal	Stable	Stable	Stable	Stable	Increase	/	Stable	Stable	Stable	Stable	Stable	Stable	Stable	/	/	Increase	Stable	Stable
Change pump	GFR increase	Wave	Stable	Stable	Stable	Stable	Stable	Increase	/	Stable	Increase	Decrease	Increase	Stable	Stable	/	/	Decrease	Slightly Decrease	Stable
Key wells	Unit aging		Stable	Stable	Stable	Stable	Stable	Gradually more than intensified 1 year	Stable	Stable	Stable	Stable	Stable	Stable	More higher	/	/	Stable	Stable	Stable
Maintenance RTU	Undercurrent protection failure		Decrease to rated value	Stable	Decrease	Decrease	Stable	Stable	/	Stable	Stable	Decrease	Stable	Stable	Stable	/	/	Stable	Stable	Stable
Maintenance RTU	Overcurrent protection failure		Increase to rated value	Stable	Decrease	Decrease	Stable	Stable	/	Stable	Stable	Decrease	Stable	Stable	Stable	/	/	Stable	Stable	Stable
Repair/Replace	Pump shaft broken		Decrease	Increase	Stable	Stable	Decrease	Increase	/	Stable	Stable	Decrease	Stable	Stable	Stable	/	/	Stable	Stable	Stable
Replace/Curing loss	Pipe loss		Decrease	Stable	Stable	Decrease	Decrease	Stable	/	Stable	Decrease	Decrease	Stable	Stable	Stable	/	/	Stable	Stable	Stable
Repair	Impeller wear		Stable	Stable	Stable	Decrease	Decrease	Increase	/	Stable	Stable	Decrease	Stable	Stable	Stable	/	/	Stable	Stable	Stable

The real-time fault diagnosis system for ESP wells employs a phased architecture, beginning with sensor-based acquisition and integration of real-time vibration, pressure, and current card data to establish a comprehensive multi-parameter database as the analytical foundation. Building upon this data infrastructure, the system implements specialized diagnostic modeling through three tailored analytical approaches: a time-frequency integrated vibration analysis model for mechanical fault detection, a Random Forest-based pressure fluctuation prediction model for flow instability identification, and a Convolutional Neural Network (CNN)-driven current card recognition model for electrical system evaluation. The system culminates with an intelligent neural network decision fusion layer, where a Gated Recurrent Unit (GRU)-enabled dynamic coupling mechanism automatically adjusts model weighting coefficients based on fault characteristics (abrupt or gradual), effectively synthesizing multi-source diagnostic outputs into unified and actionable fault conclusions. This integrated architecture enables adaptive, multi-dimensional analysis of ESP operational conditions while have a high diagnostic accuracy across varying well conditions.

Pump intake pressure and discharge pressure. The variation trends of pump intake pressure and discharge pressure serve as key indicators for analyzing well production performance. The rise and fall of intake pressure accurately reflect the formation's liquid supply capacity. Under normal ESP operating conditions, discharge pressure typically fluctuates in sync with intake pressure. Abnormal fluctuations in discharge pressure often indicate potential blockage risks in the ESP system or wellbore. By calculating the slope of intake pressure changes within specific time windows, pressure fluctuation characteristics can be quantified: When intake pressure shows sustained or sharp decline, it indicates insufficient formation energy to maintain current production rates; Conversely, stable or rising intake pressure trends demonstrate adequate formation fluid supply, suggesting potential for maintaining or increasing production.

Current and voltage of pump. Current and voltage variations typically exhibit synchronous trends. ESP manufacturers preset safety thresholds to prevent equipment damage caused by sudden current or voltage

surges. Among these electrical parameters, the changing characteristics of current can more directly reflect the operating status of the ESP and serve as critical evidence for fault diagnosis.

The ESP current card serves as a critical monitoring tool for electrical submersible pump systems by continuously recording motor current characteristics (including amplitude, fluctuation frequency, and phase patterns) to comprehensively reflect operational load and performance status. Its core functions encompass four key aspects: (1) Real-time condition monitoring - detecting current anomalies (overload, underload, or waveform distortion) to identify issues like pump blockage, gas locking, or electrical faults; (2) Predictive diagnostics - matching current signatures against failure mode databases and applying LSTM algorithms to forecast equipment degradation trends; (3) Performance optimization - comparing actual vs. design current curves to verify optimal operation at the best efficiency point (BEP) and guide VFD parameter adjustments; (4) Maintenance planning - generating service recommendations based on anomaly duration/severity while accumulating load data for remaining useful life (RUL) prediction.

Vibration. The vibration of an ESP primarily indicates whether mechanical faults have occurred, while also reflecting the fluid flow conditions inside the pump. It can detect potential blockages caused by asphaltene deposits, sand, or other obstructions. The Isolation Forest is an unsupervised anomaly detection algorithm based on a tree model. Its core principle involves randomly partitioning the feature space to quickly isolate anomalies, as shown in Fig.3. Compared to normal data, abnormal vibration signals typically exhibit characteristics such as sparsity and high volatility, making them easier to separate using isolation trees^{[17][18][19]}. ESP vibration anomalies may stem from faults such as bearing wear, rotor imbalance, or cavitation. The Isolation Forest algorithm can effectively identify these abnormal patterns without relying on historical fault-labeled data.

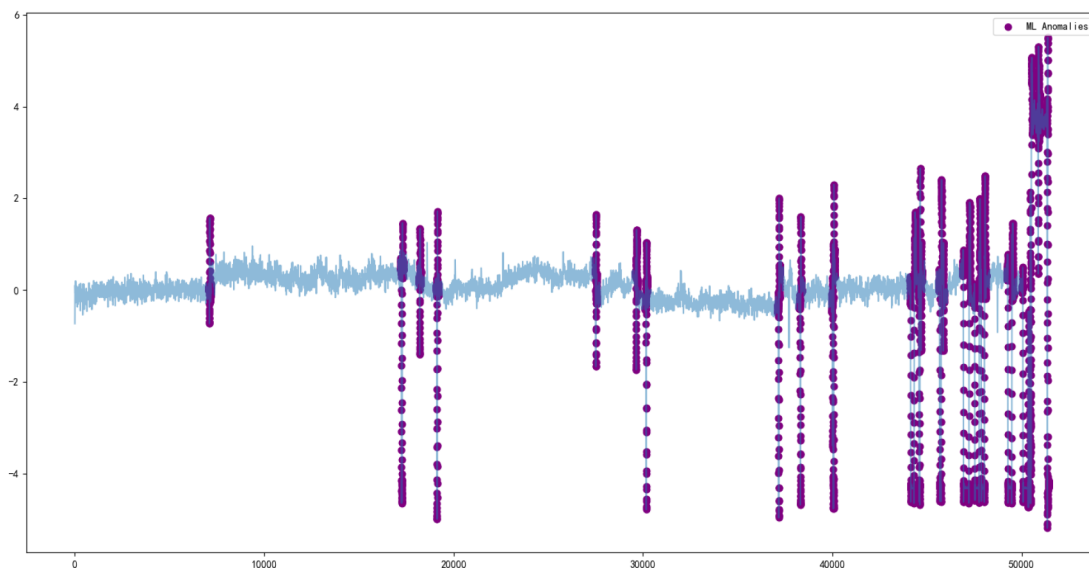


Figure 3—Vibration anomaly detection

Early warning method

At the initial ESP production stage of oilfield wells, the lack of sufficient historical data often makes purely data-driven prediction methods unreliable in terms of accuracy. To address this, a parameter-differentiated prediction strategy is adopted: for regularly varying parameters (such as current and pressure), parameter prediction and early warning methods are established; for abrupt-change parameters, parameter threshold alarm methods are employed; meanwhile, customized prediction algorithms are developed for each type of parameter. This hierarchical prediction scheme based on parameter characteristics not only overcomes the

limitations of insufficient early-stage data, but also effectively meets the needs of predictive maintenance, laying a solid foundation for subsequent model optimization as more data accumulates.

Pressure data. As previously mentioned, pump intake pressure serves as an effective indicator of reservoir pressure. By employing slope-based prediction methods to forecast pump intake pressure trends over a given period, this parameter can function as a reliable early-warning indicator for potential fluid supply insufficiency in ESP wells, as shown in Fig.4 and Fig.5.

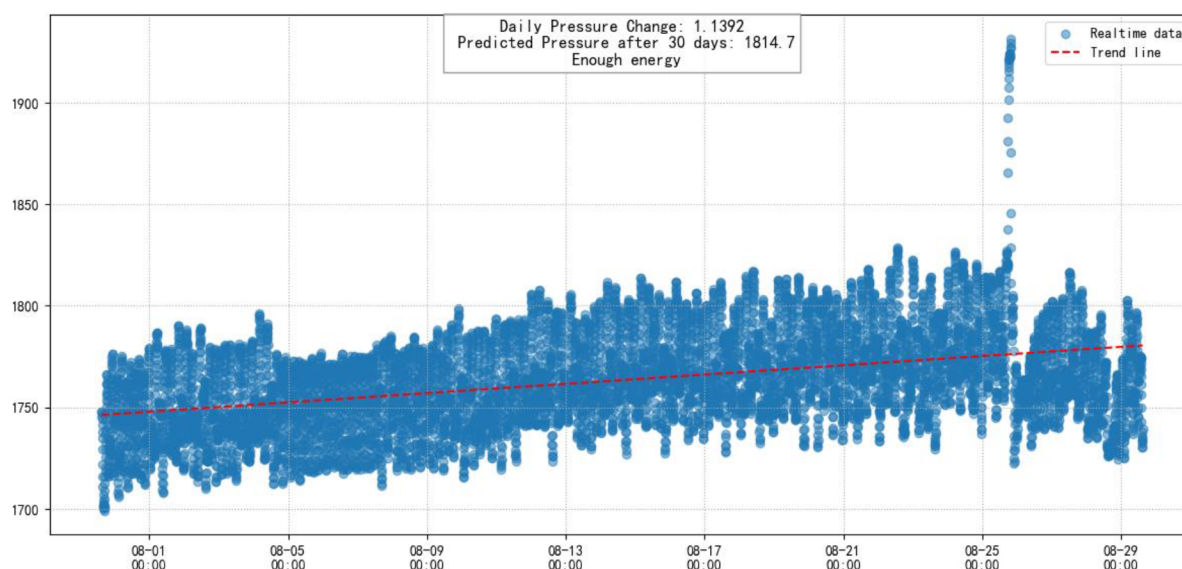


Figure 4—Identification of pump intake pressure

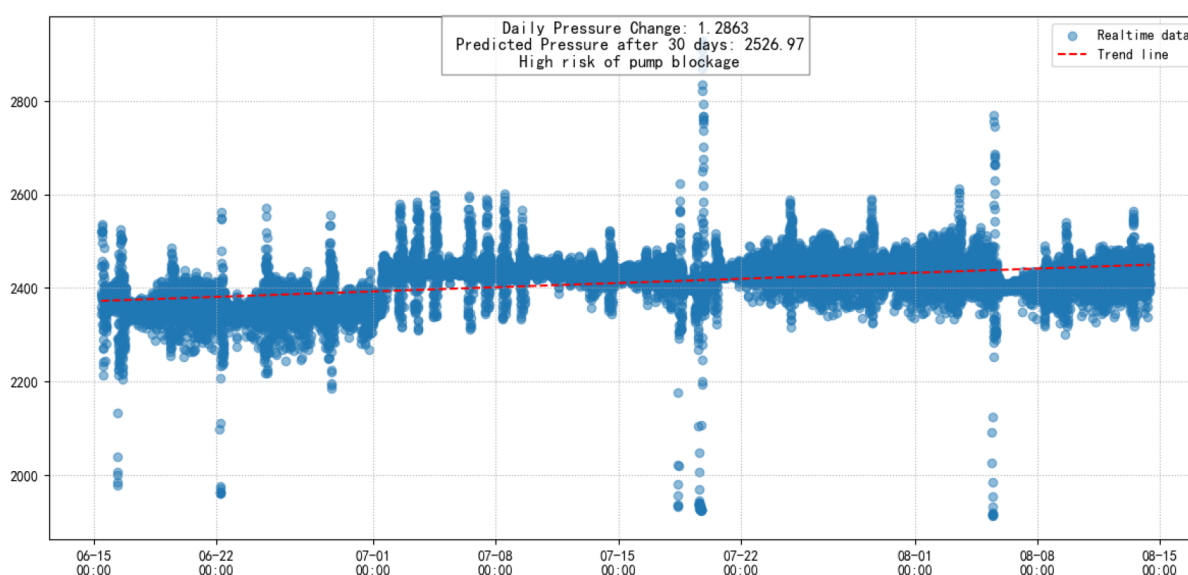


Figure 5—Identification of pump discharge pressure

Current, voltage, temperature data. This type of data typically exhibits short-term progressive changes prior to failure, followed by a sharp spike at the moment of failure. Crucially, there exists a brief pre-failure time window during which early warning becomes possible by monitoring the rate of change (slope magnitude). By analyzing this slope, can determine whether the current will reach equipment threshold levels within the imminent timeframe.

Viscosity data. Viscosity data represents one of the most unpredictable parameters in oilfield operations, primarily due to its significant dependence on reservoir fluid composition and the impracticality of real-time measurement. In field practice, viscosity monitoring for ESP well fluids is conducted through intermittent wellhead sampling followed by laboratory analysis. Given these constraints, viscosity management adopts a threshold-based approach, where exceeding predefined limits triggers immediate alarm notifications.

Integrated ESP management platform

Using the electric pump well intelligent platform to manage electric pump production is an efficient and economical management method^{[20][21][22]}. The integrated ESP well production management platform comprises the following core modules: production visualization, fault diagnosis and early warning, ESP design and production optimization, and inventory management modules.

Production visualization module. This module is designed to meet the demands of oilfield site management, with real-time data acquisition at its core. Leveraging distributed databases and cloud storage technologies, it integrates multidimensional data including well pressure, temperature, production rates, categories, and statuses. By incorporating GIS mapping for precise location tracking, the system achieves high-precision dynamic visualization of oilfield facilities and well locations(Fig. 6).



Figure 6—Home page of ESP management system

Through an interactive dashboard, managers can monitor real-time production dynamics of individual wells, quickly locate target wells by well number, and access key metrics such as operational status, liquid production, gas output, and water cut. The system homepage also provides macro-level data including daily production trends, capacity planning, and remaining production forecasts across the entire oilfield to support data-driven decision-making.

This solution establishes a closed-loop management system spanning from field operations to command dispatching, significantly enhancing global control efficiency for daily and monthly production data. By

promoting production transparency, management collaboration, and intelligent decision-making, to fully realize digital transformation objectives.

Module Workflow Description:

Log in to the system and access the homepage.

Oilfield Data Overview: i. Number of active/inactive wells, daily oil production, water cut, annual production target, and remaining production progress; ii. Layered pump inspection cycle data, including average inspection intervals and per-layer inspection cycles; iii. Planned vs. actual production, remaining production progress, water cut trends, well activation rate, and ESP contribution to total production; iv. Key well dynamics: Individual well production and water content.

Data Display in GIS map: Well locations, operational status (active/inactive), upcoming interventions, production rates, water cut, and ESP operational details.

Fault diagnose and early warning module. This module establishes an intelligent fault diagnosis and early warning system for ESP wells' real-time production status by integrating data-driven methodologies with mechanistic analysis (Fig. 7). The system performs multidimensional analysis of ESP well production conditions by retrieving required real-time operational data and static reservoir characteristic parameters from the integrated production database. Leveraging diagnostic and warning algorithms, it enables: (1) accurate fault identification, (2) proactive alerts for abnormal production states, and (3) corresponding optimization recommendations for production enhancement.

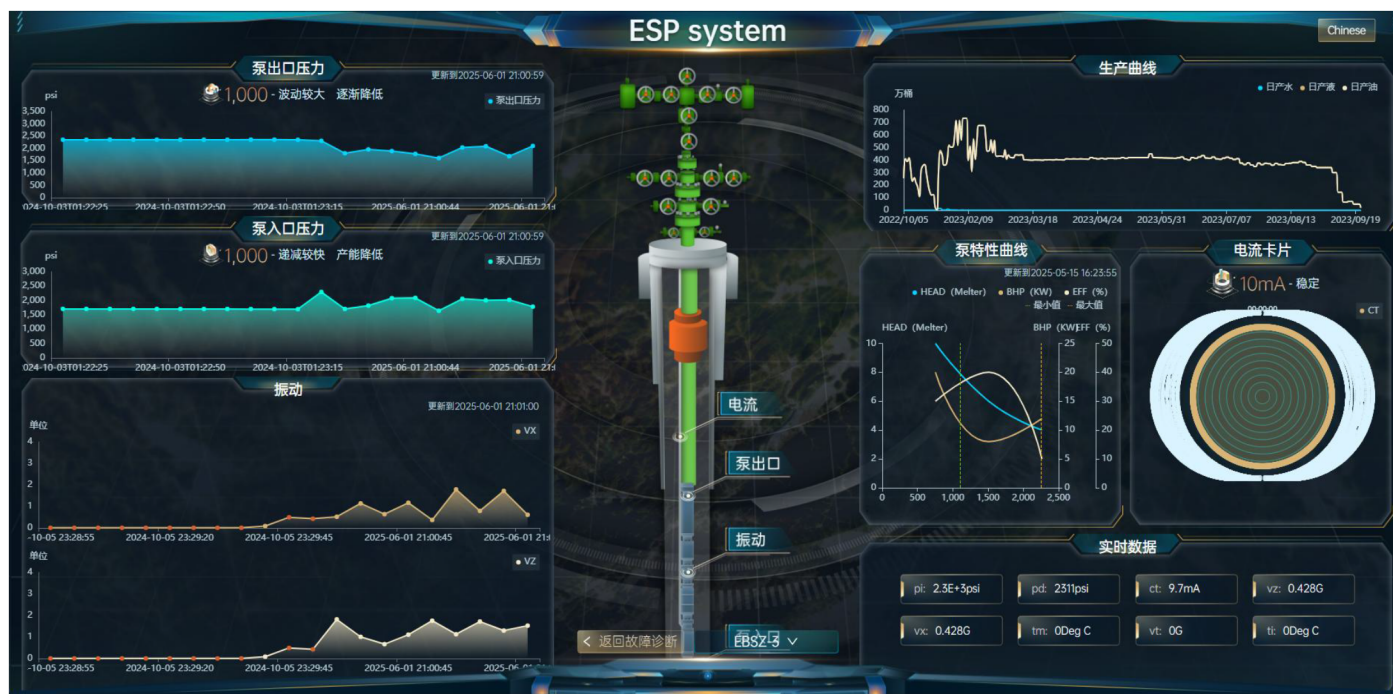


Figure 7—Diagnose and warning of ESP management system

Real-Time diagnose & warning: Dynamically tracks equipment operating status through threshold alarms and trend deviation analysis to proactively identify common failures (e.g., overcurrent, overload, pump blockage, gas locking).

Root Cause Analysis: Utilizes integrated mechanistic analysis logic table to pinpoint failure sources (e.g., pump efficiency decline caused by insufficient fluid supply), reducing misdiagnosis rates.

Predictive Maintenance: Incorporates equipment lifecycle modeling to forecast ESP remaining service life and potential failures, optimizing maintenance schedules.

Knowledge Accumulation: Builds a fault case database by annotating failure data through case reviews and expert validation, enabling continuous model iteration to improve warning accuracy.

ESP design and production optimization module. This module digitally visualizes and optimizes the ESP selection, modeling, and wellbore structure design processes, achieving automated end-to-end management from parameter invocation to dynamic simulation to enhance design efficiency and operational safety(Fig.8). The system's core functionality comprises three integrated modules - parameter library invocation, 3D wellbore structure visualization, and dynamic simulation - forming a complete intelligent solution.

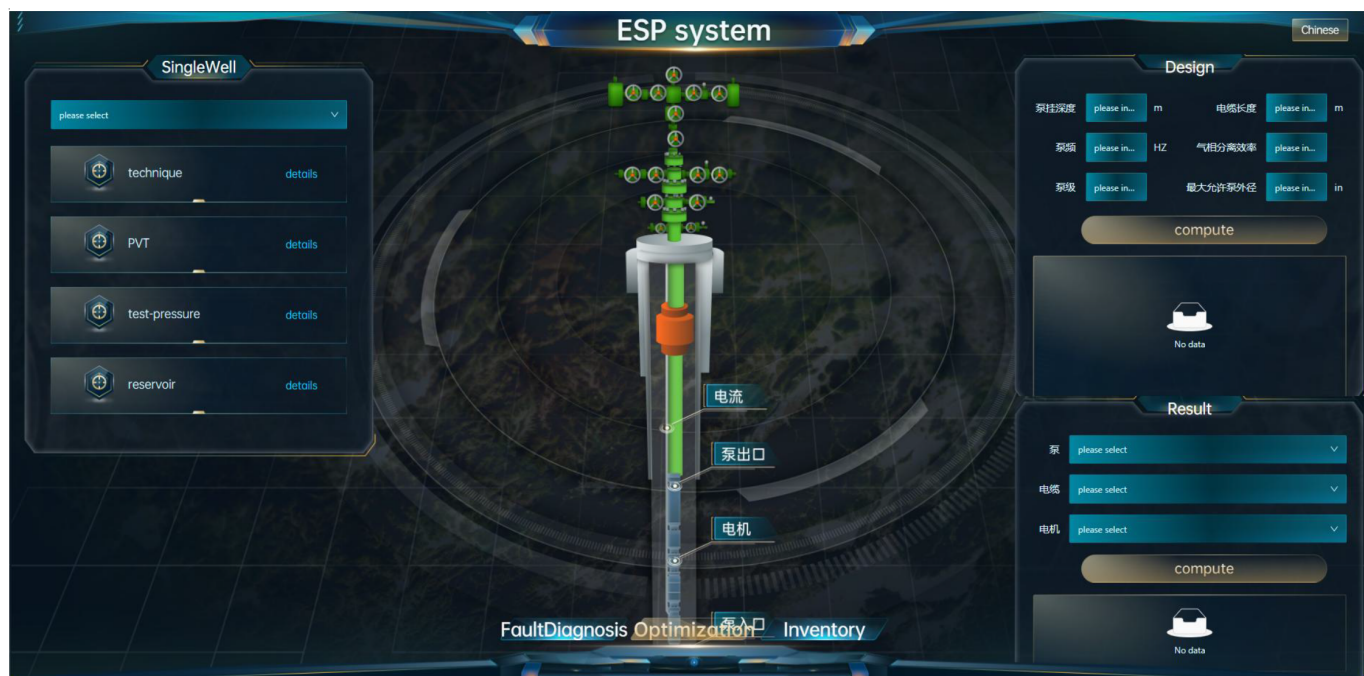


Figure 8—ESP design and production optimization of ESP management system

The parameter library module utilizes modeling software (Prosper) to automate ESP modeling and storage. Users can directly access preconfigured backend models for daily production management analysis, ensuring model accuracy while significantly reducing design cycles. Additionally, the system supports parametric modeling script development, enabling users to adjust key parameters according to operational requirements for rapid generation of customized models adapted to various working conditions. All models are saved in the database for one-click retrieval in subsequent designs, ensuring data consistency and traceability.

The wellbore structure visualization and dynamic simulation module represents the system's key innovation. Based on user-input design data, it automatically generates 3D wellbore models that intuitively display spatial configurations and relative positioning of critical components (tubing, motor, pump, cable, etc.). Through dynamic animation simulations, the ESP parameter design process becomes more visual and standardized, significantly improving modeling accuracy.

In summary, by integrating intelligent parameter library invocation, precise inventory management, and dynamic wellbore visualization, this system addresses critical pain points in conventional ESP design - including experience dependency, low efficiency, and high error rates - delivering an efficient, reliable, and intuitive digital tool for oilfield development that enhances both cost-effectiveness and operational safety.

Inventory management. The ESP inventory management module delivers real-time, visualized stock data encompassing pump models, quantities, manufacturers, and detailed technical specifications (including

rated power, applicable well depth, temperature resistance range, etc.), as shown in Fig.9. Through its filtering functionality, users can rapidly match requirements to facilitate equipment selection decisions. Furthermore, the system incorporates an intelligent inventory alert mechanism that automatically triggers notifications when stock levels fall below predefined thresholds, preventing production disruptions due to spare parts shortages. The module's integration with the parameter database ensures consistency between design-phase parameters and actual inventory specifications, thereby minimizing design discrepancies.



Figure 9—ESP design and production optimization of ESP management system

Conclusion

Field deployment of the platform has demonstrated its ability to improve ESP reliability, reduce maintenance costs, and enhance production efficiency. This study presents an integrated ESP well management platform that synergizes mechanistic analysis with data-driven modeling to optimize ESP performance and reliability. By leveraging real-time data from geological, reservoir, wellbore, and ESP systems, the platform provides a holistic approach to fault diagnosis, early warning, and production optimization.

- 1. Hybrid Diagnostic Framework: The integration of machine learning models (CNN, Random Forest, Isolation Forest) with mechanistic logic enables high-accuracy fault detection, root cause analysis, and predictive maintenance. The GRU-based neural network further enhances adaptability by dynamically adjusting model weights based on fault patterns.
- 2. Real-Time Monitoring and Early Warning: The platform’s threshold-based and trend-based warning systems proactively identify abnormalities in pressure, current, voltage, and vibration, reducing unplanned downtime.
- 3. Production Optimization: The ESP design and simulation module streamlines pump selection and wellbore configuration, while the inventory management system ensures seamless alignment between design parameters and available equipment.

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